Year 10 Higher

REVISION PACK

Sets 1

Name: _____

Class: _____

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2, 9.3)

Worked example: listing the members of a set

$P = \{x : x \text{ is a square number less than } 50\}$. List the members of set P.

Step 1	Read the rule	square numbers less than 50
Step 2	List them	$1, 4, 9, 16, 25, 36, 49$
Step 3	Answer	$\{1, 4, 9, 16, 25, 36, 49\}$

Practice: listing the members of a set

1.1 $P = \{x : x \text{ is a prime number less than } 30\}$. List the members of set P.

1.2 $P = \{x : x \text{ is a factor of } 36\}$. List the members of set P.

1.3 $P = \{x : x \text{ is a multiple of } 7 \text{ between } 10 \text{ and } 60\}$. List the members of set P.

1.4 $P = \{x : x \text{ is a cube number less than } 100\}$. List the members of set P.

1.5 $P = \{x : x \text{ is a factor of } 30\}$. List the members of set P.

1.6 $P = \{x : x \text{ is an odd number between } 20 \text{ and } 32\}$. List the members of set P.

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2, 9.3)

Worked example: intersection, union and complement

$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

A = even numbers B = factors of 10

(a) List the members of $A \cap B$. (b) Find $n(A \cup B)$. (c) Is it true that $3 \in A' \cap B'$? Give a reason.

(a)	**Step 1**	List each set	$A = \{2, 4, 6, 8, 10\},\ B = \{1, 2, 5, 10\}$
	Step 2	In both	$2, 10$
	Step 3	Answer	$\{2, 10\}$
(b)	**Step 1**	In A or B	$\{1, 2, 4, 5, 6, 8, 10\}$
	Step 2	Answer	7
(c)	**Step 1**	Is 3 in A or B?	$\text{not in } A,\ \text{not in } B$
	Step 2	Answer	$\text{Yes, 3 is in neither } A \text{ nor } B$

Practice: intersection, union and complement

1.7 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ A = multiples of 3 B = factors of 12

(a) List the members of $A \cap B$. (b) Find $n(A \cup B)$. (c) Is it true that $7 \in A' \cap B'$? Give a reason.

1.8 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ A = odd numbers B = factors of 8

(a) List the members of $A \cap B$. (b) Find $n(A \cup B)$. (c) Is it true that $6 \in A' \cap B'$? Give a reason.

1.9 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ A = multiples of 5 B = prime numbers

(a) List the members of $A \cap B$. (b) Find $n(A \cup B)$. (c) Is it true that $9 \in A' \cap B'$? Give a reason.

1.10 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ A = square numbers B = factors of 6

(a) List the members of $A \cap B$. (b) Find $n(A \cup B)$. (c) Is it true that $4 \in A' \cap B'$? Give a reason.

1.11 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ A = multiples of 4 B = factors of 8

(a) List the members of $A \cap B$. (b) Find $n(A \cup B)$. (c) Is it true that $3 \in A' \cap B'$? Give a reason.

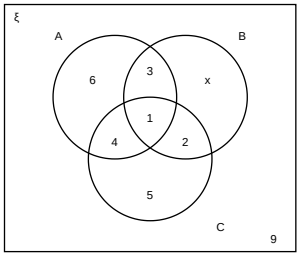
1.12 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ A = square numbers B = multiples of 3

(a) List the members of $A \cap B$. (b) Find $n(A \cup B)$. (c) Is it true that $12 \in A' \cap B'$? Give a reason.

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2, 9.3)

Worked example: three-set Venn diagrams

The Venn diagram shows the number of elements in each region. $n(\xi) = 40$

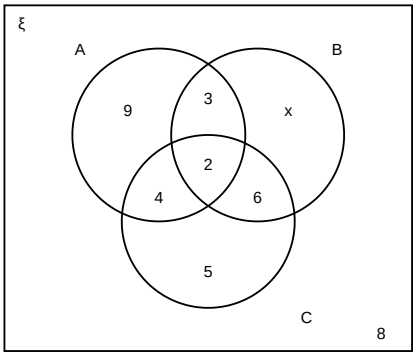


(a) Work out the value of x. (b) Find $n(A \cap B)$. (c) Find $n(C')$.

(a)	**Step 1**	Add the numbers	$6 + 5 + 3 + 4 + 2 + 1 + 9 = 30$
	Step 2	Subtract from 40	$40 - 30 = 10$
	Step 3	Answer	$x = 10$
(b)	**Step 1**	Add inside A and B	$3 + 1 = 4$
	Step 2	Answer	4
(c)	**Step 1**	Add inside C	$5 + 4 + 2 + 1 = 12$
	Step 2	Subtract from 40	$40 - 12 = 28$
	Step 3	Answer	28

Practice: three-set Venn diagrams

1.13



The Venn diagram shows the number of elements in each region.

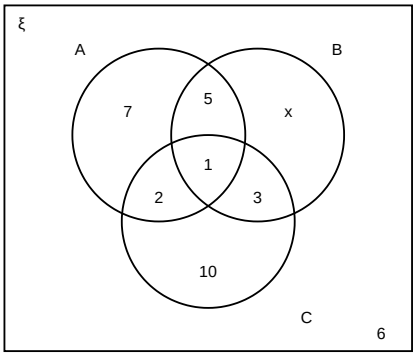
$n(\xi) = 45$

(a) Work out the value of x.

(b) Find $n(A \cap B)$.

(c) Find $n(C')$.

1.14



The Venn diagram shows the number of elements in each region.

$n(\xi) = 40$

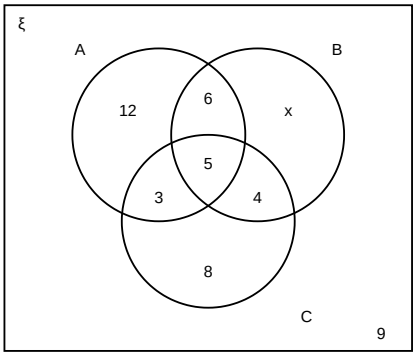
(a) Work out the value of x.

(b) Find $n(A \cap B)$.

(c) Find $n(C')$.

Practice: three-set Venn diagrams (continued)

1.15



The Venn diagram shows the number of elements in each region.

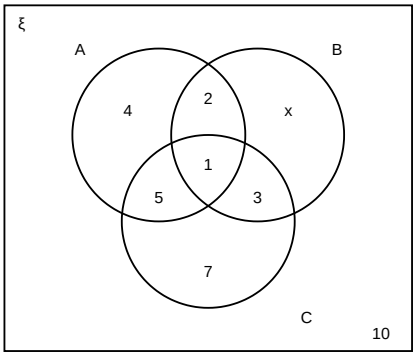
$n(\xi) = 60$

(a) Work out the value of x.

(b) Find $n(A \cap B)$.

(c) Find $n(C')$.

1.16



The Venn diagram shows the number of elements in each region.

$n(\xi) = 35$

(a) Work out the value of x.

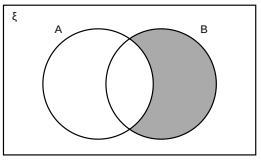
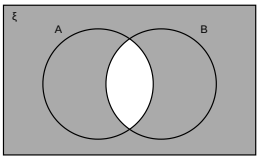
(b) Find $n(A \cap B)$.

(c) Find $n(C')$.

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2, 9.3)

Worked example: shading regions

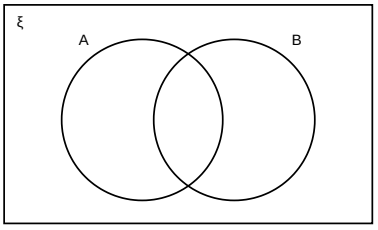
Shade the region on each Venn diagram. (a) $A' \cap B$ (b) $(A \cap B)'$

(a)	**Step 1**	A': outside A	
	Step 2	$\cap B$: and inside B	
	Step 3	Shade B only	
(b)	**Step 1**	$A \cap B$: the middle	
	Step 2	$'$: everything else	
	Step 3	Shade all but the middle	

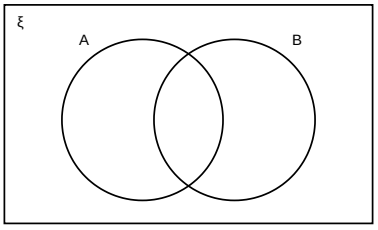
Practice: shading regions

1.17 Shade the region on each Venn diagram.

(a) $A \cup B'$

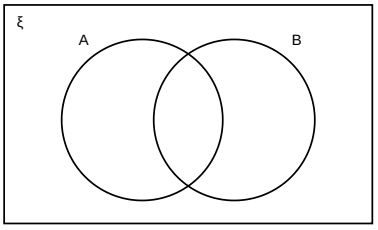


(b) A'

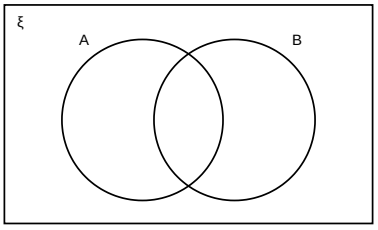


1.18 Shade the region on each Venn diagram.

(a) B'

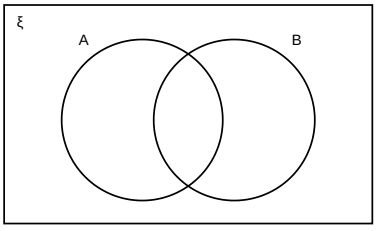


(b) $A \cap B$

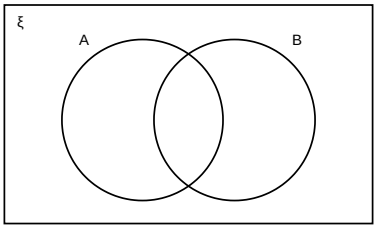


1.19 Shade the region on each Venn diagram.

(a) $A' \cup B$



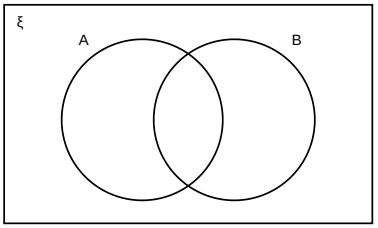
(b) $A \cup B$



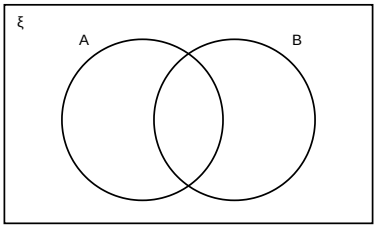
Practice: shading regions (continued)

1.20 Shade the region on each Venn diagram.

(a) $(A \cap B)'$

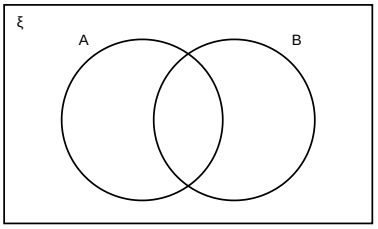


(b) B'

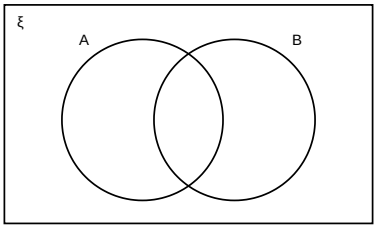


1.21 Shade the region on each Venn diagram.

(a) $A' \cap B$

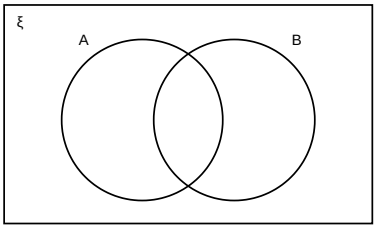


(b) $A \cup B'$

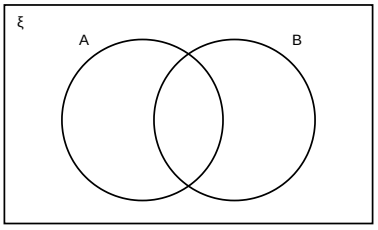


1.22 Shade the region on each Venn diagram.

(a) A'



(b) $A' \cup B$



Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2, 9.3)

Worked example: problems with two sets

30 students were asked if they like tea or coffee. 18 like tea. 15 like coffee. 4 like neither.

(a) How many students like both tea and coffee? (b) How many students like tea but not coffee?

(a)	**Step 1**	Add	$18 + 15 + 4 = 37$
	Step 2	Over 30 = counted twice (both)	$37 - 30 = 7$
	Step 3	Answer	7
(b)	**Step 1**	Tea take away both	$18 - 7 = 11$
	Step 2	Answer	11

Practice: problems with two sets

1.23 40 students were asked if they like tea or coffee. 22 like tea. 16 like coffee. 6 like neither.

(a) How many students like both tea and coffee? (b) How many students like tea but not coffee?

1.24 50 students were asked if they like swimming or basketball. 28 like swimming. 25 like basketball. 5 like neither.

(a) How many students like both swimming and basketball? (b) How many students like swimming but not basketball?

1.25 35 students were asked if they like maths or art. 20 like maths. 14 like art. 8 like neither.

(a) How many students like both maths and art? (b) How many students like maths but not art?

1.26 45 students were asked if they like cats or dogs. 25 like cats. 20 like dogs. 10 like neither.

(a) How many students like both cats and dogs? (b) How many students like cats but not dogs?

1.27 30 students were asked if they like apples or bananas. 16 like apples. 12 like bananas. 6 like neither.

(a) How many students like both apples and bananas? (b) How many students like apples but not bananas?

1.28 60 students were asked if they like football or cricket. 35 like football. 30 like cricket. 5 like neither.

(a) How many students like both football and cricket? (b) How many students like football but not cricket?